High Dimensional Simplicity

Chase Pittman and Elena Ruiz

Advisor: Dr. Adam Cartisano

Belmont University

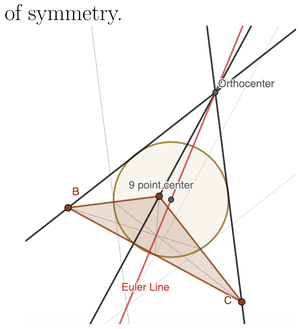
Introduction

Motivating Question: Can any triangle be realized as the image of a projected equilateral triangle? Furthermore, can any orthocentric system be realized as the projected image of the vertices of a tetrahedron?



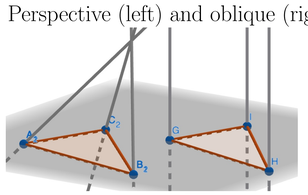
In a geometry software like GeoGebra, we draw an arbitrary triangle, it's Euler line, and its altitudes. As we move the vertices around, changing the triangle, we notice that our drawing begins to look like a 3-dimensional structure, despite being 2-dimensional. The triangle's vertices and it's orthocenter appear to be the 4 vertices of a tetrahedron, and the Euler line appears to be an axis of symmetry.

Furthermore, we draw the nine- point circle of our original triangle and find that the nine-point center sits in the expected location of the tetrahedron's centroid. The question is, are these intuitions justified?



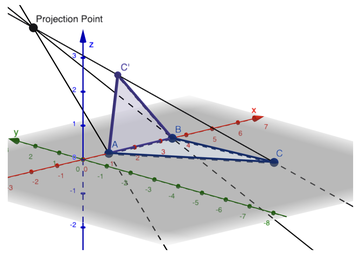
Terminology

- **Centroid:** The point of concurrency of the medians, also the average of the vertices.
- **Orthocenter:** The point of concurrency of the altitudes.
- **Nine Point Center:** Center of the nine point circle, also the average of the three vertices and the orthocenter.
- **Euler Line:** The line spanned by the nine point center, centroid, circumcenter, and orthocenter in any triangle.
- **Orthocentric System:** The set of all three vertices of a given triangle and the triangle's orthocenter. Of these four points, any three can be chosen to be the vertices of a triangle, and the fourth will always be that triangle's orthocenter, yielding four different triangles and their orthocenters for each orthocentric system.
- **Types of Projections:** Perspective (left) and oblique (right)



Preliminary Results

Theorem 1.1 Every triangle in $\mathbb{R}^2$ is the projection of an equilateral triangle in $\mathbb{R}^3$.



Theorem 1.2 Any triangle can be lifted to another in perspective while projecting centroids to centroids, on the condition that the the triangles are similar and their spanning planes are parallel.

Proof Summary: Using the properties of similar triangles and conditions for linear independence of vectors, it can be shown that the spanning planes containing these triangles must be the same constant apart, making them parallel, and that the sides of these triangles must be some constant multiple of one another. From these relationships, it can be shown that in this case, and no other, the centroids of the triangles must project to one another.

Main Result

Lemma 2.1 Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Let $S = \{\vec{v}_1, \ldots, \vec{v}_k\}$ be a set of k vectors in $\mathbb{R}^n$. Then the centroid of S maps to the centroid of the images of the vectors in S under T.

Proof: $$T\left(\frac{1}{k}\sum_{i=1}^{k}\vec{v}_i\right) = \frac{1}{k}T\left(\sum_{i=1}^{k}\vec{v}_i\right) = \frac{1}{k}\sum_{i=1}^{k}T\left(\vec{v}_i\right).$$ □

Theorem 2.2 (Pohlke-Schwarz Theorem [1]**)** Every quadrilateral can be viewed as the oblique projection of a tetrahedron similar to a given one.

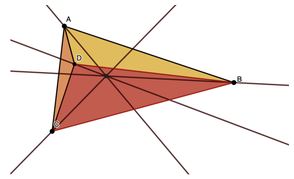
Theorem 2.3 Every orthocentric system is the oblique projection of a regular tetrahedron in $\mathbb{R}^3$. Moreover, the Euler lines of the four triangular subsets of the orthocentric system are the projections of the axes of symmetry of the tetrahedron.

Proof: The first claim follows directly from Theorem 2.2, where the quadrilateral is our starting orthocentric system, and the given tetrahedron is regular. To prove the second claim, let $\triangle ABC$ be one of the four triangular subsets of the orthocentric system. By Lemma 2.1, the centroid of an equilateral face projects to the centroid of $\triangle ABC$. The fourth point of the tetrahedron projects to the fourth point of the orthocentric system. By the orthocentric system property, this is the orthocenter of $\triangle ABC$.
Then the line joining the centroid of an equilateral face of the regular tetrahedron and its opposite vertex projects to the line joining the centroid and orthocenter of $\triangle ABC$. The former is just an axis of symmetry in the tetrahedron, and the latter is the Euler line of $\triangle ABC$. □

Discussion

Here are some thoughts to provide context for our results:

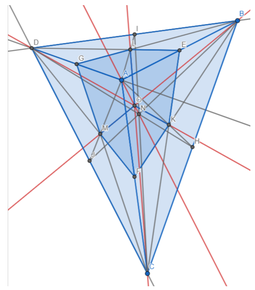
1. The proof of the Pohlke-Schwarz depends on the Fundamental Theorem of Axonomoetry[2] which states that three arbitrary line segments, originating from a common point, which are not contained in a line, can be considered the oblique projection of three edges of a cube.
2. The four Euler lines of an arbitrary orthocentric system are concurrent at it's nine point center. When lifted to a tetrahedron in $\mathbb{R}^3$, the 4 axes of symmetry of the tetrahedron intersect at it's centroid.
   Corollary to Theorem 2.3: The centroid of the regular tetrahedron projects to the nine-point center.



3. Our guiding objective throughout this project is simplification: it is often easier to study simpler objects, and so finding a way to view complex shapes in lower dimensions as less complex shapes in higher dimensions offers a natural way to view some of these familiar objects.

Future Work

Pictured to the left are several structures which arise naturally just by observing the orthocentric system in GeoGebra.



Three of the four points in the orthocentric system form an acute triangle (convex hull). Those three points can be seen as the vanishing points for a perspective drawing of a cube, pictured below. Several collinearity theorems can be derived from this approach. One can also hunt for additional structures which arise from the triangle (We have notices what appear to be a hypercube and a Fano plane.

Is the pictured cube related to the Fundamental Theorem of Axonometry?

References

1. Renato Manfrin. "Addendum to Pohlke's Theorem, a Proof of Pohlke-Schwarz's Theorem." *Journal for Geometry and Graphics*, vol. 23, no. 1, 2019, pp. 41–44. Heldermann Verlag, https://www.heldermann-verlag.de/jgg/jgg23/j23h1manf.pdf.
2. Renato Manfrin. "A Proof of Pohlke's Theorem with an Analytic Determination of the Reference Trihedron." *Journal for Geometry and Graphics*, vol. 22, no. 2, 2018, pp. 195-205, https://www.heldermann-verlag.de/jgg/jgg22/j22h2manf.pdf.

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